DATA 211 – Introduction to Data Science
Topic 1: Introduction to Data Science

Question bank: 40 multiple choice + 10 programming

Questions from DATA 211 / ISE 291 old exams and quizzes (AI_Work). **[new]** marks new questions written in the same style.

Part A: Multiple choice

Q1. [new]

According to the lecture, "Big Data" refers to data that is:

(A) small enough to be stored in a single spreadsheet

(B) too large and too complex for conventional data tools to capture, store and analyze

(C) always stored in relational databases

(D) collected only by social media companies

Answer: (B). Big data is data too large and too complex for conventional tools to capture, store and analyze.

My answer

Q2. [new]

The 3V's of Big Data presented in the lecture are:

(A) Volume, Variety, Velocity

(B) Volume, Value, Visualization

(C) Validity, Variety, Veracity

(D) Velocity, Vectorization, Volume

Answer: (A). The lecture lists Volume, Variety and Velocity.

My answer

Q3. [new]

Which "V" of big data describes the sheer amount of data that is generated and stored (for example, 10 terabytes of data from a single flight)?

(A) Velocity

(B) Volume

(C) Variety

(D) Veracity

Answer: (B). Volume is the amount (size) of the data.

My answer

Q4. [new]

About 7 billion shares are traded on US stock markets each day, and the trades must be captured and processed as they happen. This aspect, the speed at which data is generated and must be processed, is called:

(A) Variety

(B) Volume

(C) Velocity

(D) Visualization

Answer: (C). Velocity is the speed at which data arrives and must be processed.

My answer

Q5. [new]

A company stores customer tweets (text), product photos (images), sensor readings (numbers) and call recordings (audio). Which "V" of big data does this mainly illustrate?

(A) Volume

(B) Velocity

(C) Variety

(D) Value

Answer: (C). Many different kinds/formats of data is Variety.

My answer

Q6. [new]

According to the lecture, approximately what percentage of the world's data was generated in the last two years?

(A) 10%

(B) 50%

(C) 75%

(D) 90%

Answer: (D). The slide states that 90% of the world's data was generated in the last two years.

My answer

Q7. [new]

According to the lecture, approximately how much data is generated in one flight from New York to London?

(A) 10 Megabytes

(B) 10 Gigabytes

(C) 10 Terabytes

(D) 10 Petabytes

Answer: (C). The slide gives 10 Terabytes per flight.

My answer

Q8. [new]

The lecture states: "Data on its own is useless unless you can make sense of it!" Which statement best reflects this idea?

(A) Raw data becomes valuable only when it is analyzed and turned into meaningful insight

(B) The more data you store, the more value you automatically get

(C) Data should be deleted after it is collected to save storage

(D) Only numerical data can ever be useful

Answer: (A). Value comes from making sense of (analyzing) the data, not from storing it.

My answer

Q9. TOPIC 2 & 3 (ISE 291 Sec 4, Quiz 1)

What is data?

(A) The result of processing information

(B) The raw material from which information is obtained

(C) The process of converting information into a usable format

(D) The process of analyzing information to extract insights

Answer: (B). Data is the raw material; information is obtained by processing it.

My answer

Q10. 223 Midterm

Data scientists focus more on producing _____ products than answering operational corporate questions.

(A) Blind spot

(B) Data-driven

(C) Experience-based

(D) Knowledge-based

Answer: (B). Data scientists build data-driven products.

My answer

Q11. [new]

In the machine learning approach described in the lecture, the programmer:

(A) writes an explicit if-else rule for every possible case

(B) programs an algorithm that automatically learns from previous data

(C) manually labels every future data point

(D) stores the data without analyzing it

Answer: (B). Machine learning: program an algorithm to automatically learn from previous data.

My answer

Q12. [new]

An e-mail spam filter improves by learning from thousands of past e-mails that were marked as "spam" or "not spam". This is an example of:

(A) machine learning

(B) data storage only

(C) manual rule-based programming

(D) data entry

Answer: (A). The algorithm learns automatically from previous (labelled) data.

My answer

Q13. [new]

In the course content slide, which pair is listed under "Big data concepts"?

(A) K-means and Hierarchical

(B) MapReduce and Hadoop

(C) Variance and Covariance

(D) PCA and Gradient Descent

Answer: (B). Big data concepts: MapReduce, Hadoop.

My answer

Q14. [new]

Python was started as a hobby project by:

(A) Dennis Ritchie

(B) Guido van Rossum

(C) James Gosling

(D) Bjarne Stroustrup

Answer: (B). Python was started by Guido van Rossum as a hobby.

My answer

Q15. [new]

Which of the following is NOT a property of Python given in the lecture?

(A) It is open source

(B) It is free

(C) It is versatile

(D) It requires a paid license for data analysis

Answer: (D). Python is open source, free and versatile.

My answer

Q16. [new]

According to the lecture, Python today is one of the most important languages for:

(A) data science, machine learning and general software development

(B) styling web pages only

(C) writing operating system kernels only

(D) spreadsheet formulas only

Answer: (A). It has a large scientific computing and data analysis community.

My answer

Q17. [new]

Which Python library introduces objects for multidimensional arrays and matrices, together with functions to easily perform advanced mathematical and statistical operations on them?

(A) NumPy

(B) Pandas

(C) Seaborn

(D) Scikit-learn

Answer: (A). This is the lecture's description of NumPy.

My answer

Q18. [new]

The lecture states that "many other Python libraries are built on" which library?

(A) matplotlib

(B) NumPy

(C) Seaborn

(D) Jupyter

Answer: (B). SciPy, Pandas, scikit-learn and others are built on NumPy.

My answer

Q19. T253 Quiz 1 (Sec 04/05)

What are vectorized operations?

(A) Operations that can be done on lists with loops.

(B) Operations that can be done on arrays with loops.

(C) Operations that can be done on lists without loops.

(D) Operations that can be done on arrays without loops.

Answer: (D). Vectorization applies an operation to a whole array at once, with no explicit loop.

My answer

Q20. Quiz 1 (T253, Version A)

Which of the following is true about vectorization in NumPy?

(A) It only works on 1D arrays.

(B) It converts arrays into vectors.

(C) It only works with numeric arrays.

(D) It avoids explicit Python loops.

Answer: (D). Vectorized operations work on whole arrays of any dimension without explicit loops.

My answer

Q21. [new]

According to the lecture, what is the main benefit of NumPy's vectorization of mathematical operations on arrays and matrices?

(A) It significantly improves performance

(B) It makes arrays immutable

(C) It converts arrays into data frames

(D) It draws plots automatically

Answer: (A). Vectorization significantly improves performance.

My answer

Q22. [new]

What is the output of the following code?

```
import numpy as np
a = [1, 2, 3]
b = np.array([1, 2, 3])
print(a * 2)
print(b * 2)
```

(A) `[2, 4, 6]` then `[2 4 6]`

(B) `[1, 2, 3, 1, 2, 3]` then `[2 4 6]`

(C) `[1, 2, 3, 1, 2, 3]` then `[1 2 3 1 2 3]`

(D) Error then `[2 4 6]`

Answer: (B). A list times 2 repeats the list; a NumPy array times 2 is element-wise (vectorized).

My answer

Q23. [new]

Which line is the vectorized NumPy equivalent of the loop below?

```
prices = [10, 20, 30]
new = []
for p in prices:
    new.append(p * 1.15)
```

(A) `new = np.array(prices) * 1.15`

(B) `new = prices * 1.15`

(C) `new = [prices] * 1.15`

(D) `new = np.array(prices * 1.15)`

Answer: (A). Convert to an array first, then multiply element-wise; (B) and (D) raise a TypeError (list times float).

My answer

Q24. [new]

Which library is a collection of algorithms for linear algebra, differential equations, numerical integration, optimization and statistics?

(A) SciPy

(B) Pandas

(C) matplotlib

(D) Seaborn

Answer: (A). This is the lecture's description of SciPy.

My answer

Q25. [new]

Which statement about SciPy is TRUE according to the lecture?

(A) It is built on NumPy and is part of the SciPy Stack

(B) It is a replacement for the Python language

(C) It is used only for plotting

(D) It is built on scikit-learn

Answer: (A). SciPy is part of the SciPy Stack and is built on NumPy.

My answer

Q26. TOPIC 4 & 5 (ISE 291 Sec 4, Quiz 2)

Which library in Python is commonly used for Data Frames?

(A) Pandas

(B) Numpy

(C) Matplotlib

(D) Scikit-learn

Answer: (A). Pandas provides the Series and DataFrame structures.

My answer

Q27. [new]

Pandas adds data structures for table-like data that are similar to those in R. These structures are:

(A) Series and Data Frames

(B) Tensors and Graphs

(C) Figures and Axes

(D) Lists and Tuples

Answer: (A). Pandas provides Series and DataFrames (similar to R).

My answer

Q28. [new]

You have a table of customer records with some missing values that must be merged with a second table and then sorted. Which library provides tools designed for this?

(A) matplotlib

(B) Seaborn

(C) Pandas

(D) SciPy

Answer: (C). Pandas handles merging, sorting and missing data in table-like data.

My answer

Q29. [new]

Which of the following is NOT listed in the lecture as a Pandas data-manipulation tool?

(A) Merging

(B) Aggregation

(C) Handling missing data

(D) Solving differential equations

Answer: (D). Differential equations belong to SciPy; Pandas does reshaping, merging, sorting, slicing, aggregation and missing data.

My answer

Q30. [new]

Which Python library is a 2D plotting library that produces publication-quality figures and has a set of functionalities similar to MATLAB?

(A) NumPy

(B) matplotlib

(C) Pandas

(D) Scikit-learn

Answer: (B). This is the lecture's description of matplotlib.

My answer

Q31. [new]

According to the lecture, which kinds of output can matplotlib produce?

(A) Line plots, scatter plots, bar charts, histograms and pie charts

(B) Only line plots

(C) Data frames and series

(D) Trained classification models

Answer: (A). matplotlib draws line plots, scatter plots, bar charts, histograms, pie charts, etc.

My answer

Q32. [new]

Which set of imports uses the standard aliases for the data science libraries introduced in the lecture?

(A) `import numpy as pd`, `import pandas as np`, `import matplotlib as sns`

(B) `import numpy as np`, `import pandas as pd`, `import matplotlib.pyplot as plt`

(C) `import numpy.pyplot as plt`, `import pandas as pd`, `import seaborn as np`

(D) `import NumPy`, `import Pandas`, `import Matplotlib`

Answer: (B). `np`, `pd` and `plt` are the standard aliases; (D) fails because module names are lower-case.

My answer

Q33. TOPIC 4 & 5 (ISE 291 Sec 4, Quiz 2)

Which library provides a variety of plots such as histogram, boxplots, lineplots, scatterplots, countplot, violinplot, swarmplot, pairplot, catplot etc.?

(A) Matplotlib

(B) Seaborn

(C) Pandas

(D) Scipy

Answer: (B). countplot, violinplot, swarmplot, pairplot and catplot are Seaborn functions.

My answer

Q34. **[new]**

According to the lecture, Scikit-learn provides:

(A) web development frameworks

(B) machine learning algorithms for classification, regression, clustering and model validation

(C) spreadsheet formatting tools

(D) only plotting functions

Answer: (B). Scikit-learn is the machine learning library.

My answer

Q35. **[new]**

Scikit-learn is built on which libraries?

(A) NumPy, SciPy and matplotlib

(B) Pandas only

(C) Seaborn and Jupyter

(D) R and MATLAB

Answer: (A). Scikit-learn is built on NumPy, SciPy and matplotlib.

My answer

Q36. [new]

A data scientist wants to (i) store and manipulate a table of customer records and (ii) fit a regression model to it. Which pair of libraries is most appropriate?

(A) matplotlib and Seaborn

(B) Pandas and Scikit-learn

(C) SciPy and matplotlib

(D) Seaborn and matplotlib

Answer: (B). Pandas for table-like data, Scikit-learn for machine learning (regression).

My answer

Q37. [new]

Which of the following is NOT a feature of Jupyter notebooks (Noteable) mentioned in the lecture?

(A) Web-based environment

(B) Combines both text and code in one document

(C) Comes with a great number of useful packages

(D) Can only run compiled C programs

Answer: (D). Jupyter is an easy, web-based environment that mixes text and Python code.

My answer

Q38. [new]

Which of the following is NOT one of the course objectives?

(A) Understand and prepare data

(B) Analyze and interpret data

(C) Apply and validate data science techniques

(D) Design computer hardware for data centers

Answer: (D). The three objectives are understand/prepare, analyze/interpret, and apply/validate.

My answer

Q39. [new]

According to the learning outcomes, the course focuses on analyzing and processing which kind of data?

(A) Numerical and categorical structured data

(B) Only unstructured audio and video data

(C) Only image data

(D) Only free-text documents

Answer: (A). Outcome: analyze and process numerical and categorical structured data.

My answer

Q40. [new]

Which course topic is marked as "Not Part of Exam"?

(A) Basic Statistics

(B) Exploratory Data Analysis

(C) Database, SQL & Big Data

(D) Linear Regression Methods

Answer: (C). Database, SQL & Big Data is not part of the exam.

My answer

Part B: Programming

Q41. [new]

Python's basic data types include `int`, `float`, `str`, `list` and `dict`. Given

```
values = [42, 3.14, "KFUPM", [1, 2, 3], {"course": "DATA 211"}, True, 7, "Python"]
```

print each value together with the name of its type, then build and print a dictionary that counts how many values there are of each type.

Answer:

```
values = [42, 3.14, "KFUPM", [1, 2, 3], {"course": "DATA 211"}, True, 7, "Python"]
counts = {}
for v in values:
    name = type(v).__name__
    print(repr(v), "->", name)
    counts[name] = counts.get(name, 0) + 1
print(counts)
# {'int': 2, 'float': 1, 'str': 2, 'list': 1, 'dict': 1, 'bool': 1}
```

My answer

Q42. [new]

Structured data is organized in rows and columns. The student records below are stored as a list of dictionaries.

```
students = [
    {"id": 101, "name": "Ali",  "major": "DATA", "gpa": 3.4},
    {"id": 102, "name": "Sara", "major": "ICS",  "gpa": 3.8},
    {"id": 103, "name": "Omar", "major": "DATA", "gpa": 2.9},
    {"id": 104, "name": "Huda", "major": "MATH", "gpa": 3.6},
]
```

Convert them into a Pandas DataFrame, then print (a) its shape, (b) its column names, (c) the data type of each column, (d) the mean GPA, and (e) the rows of students with GPA above 3.5.

Answer:

```
import pandas as pd
df = pd.DataFrame(students)
print(df.shape)                 # (4, 4)
print(list(df.columns))         # ['id', 'name', 'major', 'gpa']
print(df.dtypes)
```

```
print(df["gpa"].mean())          # 3.425
print(df[df["gpa"] > 3.5])       # Sara and Huda
```

My answer

Q43. [new]

Tweets are an example of unstructured text. Turn the tweets below into a structured DataFrame with the columns `user` (without the `@`), `n_words` (number of words after the colon), `hashtags` (list of words starting with `#`) and `n_hashtags`.

```
tweets = [
    "@ali: Loving #DataScience and #Python today",
    "@sara: Big data is everywhere #BigData",
    "@omar: Just finished my first notebook",
]
```

Answer:

```
import pandas as pd
rows = []
for t in tweets:
    user, text = t.split(":", 1)
    words = text.split()
    tags = [w for w in words if w.startswith("#")]
    rows.append({"user": user.lstrip("@"), "n_words": len(words),
                 "hashtags": tags, "n_hashtags": len(tags)})
tw = pd.DataFrame(rows)
print(tw)
#    user  n_words                 hashtags  n_hashtags
# 0   ali        5  [#DataScience, #Python]           2
# 1  sara        5               [#BigData]           1
# 2  omar        5                       []           0
```

My answer

Q44. [new]

A data team has the files below. Tables such as `.csv` and `.xlsx` files are structured; images, audio and video are unstructured. Write code that builds a dictionary mapping each file name to `"structured"` or `"unstructured"` using its extension, and prints how many files are in each group.

```
files = ["sales.csv", "photo.jpg", "grades.xlsx", "lecture.mp4",
         "survey.csv", "voice_note.mp3", "students.xlsx", "scan.png"]
```

Answer:

```
structured_ext = {"csv", "xlsx"}
kind = {}
for f in files:
    ext = f.rsplit(".", 1)[1]
    kind[f] = "structured" if ext in structured_ext else "unstructured"
print(kind)
n_struct = sum(1 for k in kind.values() if k == "structured")
print("structured:", n_struct, "unstructured:", len(files) - n_struct)
# structured: 4 unstructured: 4
```

My answer

Q45. [new]

The lecture's big data figures are: 7 billion shares traded per day, 400 million tweets per day and 3 billion Facebook likes per day. Using NumPy arrays (no loops for the arithmetic), compute how many

of each happen per second and per year (365 days). Also convert the 10 TB generated in one flight into GB and MB (use 1 TB = 1000 GB = 1,000,000 MB).

Answer:

```
import numpy as np
names = np.array(["shares traded", "tweets", "Facebook likes"])
per_day = np.array([7e9, 400e6, 3e9])
per_second = per_day / (24 * 60 * 60)   # vectorized
per_year = per_day * 365                # vectorized
for n, s, y in zip(names, per_second, per_year):
    print(f"{n}: {s:,.0f} per second, {y:.3e} per year")
# shares traded: 81,019 per second, 2.555e+12 per year
# tweets: 4,630 per second, 1.460e+11 per year
# Facebook likes: 34,722 per second, 1.095e+12 per year
tb = 10
print("GB:", tb * 1000, "MB:", tb * 1000**2)   # GB: 10000 MB: 10000000
```

My answer

Q46. [new]

Show that NumPy vectorization gives the same result as a loop but is faster. For the integers 1 to 1,000,000, compute the sum of their squares (i) with a plain Python `for` loop and (ii) with a vectorized NumPy expression. Print whether the two results are equal and the time each method took (use the `time` module).

Answer:

```
import numpy as np
import time
x = np.arange(1, 1_000_001)

t0 = time.time()
total_loop = 0
for v in x.tolist():
    total_loop += v * v
t1 = time.time()
total_vec = int(np.sum(x.astype(np.int64) ** 2))
t2 = time.time()

print(total_loop == total_vec, total_vec)   # True 333333833333500000
print(f"loop: {t1 - t0:.4f} s, vectorized: {t2 - t1:.4f} s")
```

My answer

Q47. [new]

Pandas allows handling missing data. Using the data below, (a) count the missing values in each column, (b) fill missing `quiz` values with the column mean and missing `major` values with `"Unknown"`, and (c) print the table sorted by `quiz` from highest to lowest.

```
data = {"name": ["Ali", "Sara", "Omar", "Huda", "Fahad"],
        "quiz": [8.0, None, 6.5, 9.0, None],
        "major": ["DATA", "ICS", None, "DATA", "MATH"]}
```

Answer:

```
import pandas as pd
df = pd.DataFrame(data)
print(df.isnull().sum())        # name 0, quiz 2, major 1
df["quiz"] = df["quiz"].fillna(df["quiz"].mean())   # mean = 7.8333
df["major"] = df["major"].fillna("Unknown")
print(df.sort_values("quiz", ascending=False))
```

My answer

Q48. [new]

Use matplotlib to draw a bar chart of the daily big data figures from the lecture: shares traded (7 billion), tweets (400 million) and Facebook likes (3 billion). Show the values in billions, label the y-axis "Billions per day", add a title, and save the figure as `bigdata.png`.

Answer:

```
import matplotlib.pyplot as plt
labels = ["Shares traded", "Tweets", "Facebook likes"]
values = [7e9, 400e6, 3e9]
plt.figure(figsize=(6, 4))
plt.bar(labels, [v / 1e9 for v in values])
plt.ylabel("Billions per day")
plt.title("Big data generated each day")
plt.savefig("bigdata.png")
plt.show()
```

My answer

Q49. [new]

The course grading weights (in %) and one student's scores (out of 100) are stored in dictionaries:

```
weights = {"Homework": 5, "Quizzes": 10, "Project": 20,
           "Midterm": 20, "Final": 40, "Participation": 5}
scores = {"Homework": 90, "Quizzes": 75, "Project": 88,
          "Midterm": 70, "Final": 82, "Participation": 100}
```

Write code that (a) checks that the weights sum to 100, (b) computes the student's weighted total, and (c) prints the assessment with the largest weight.

Answer:

```
print(sum(weights.values()) == 100)      # True
total = 0
for item, w in weights.items():
    total += scores[item] * w / 100
print(round(total, 2))                   # 81.4
largest = max(weights, key=weights.get)
print(largest)                           # Final
```

My answer

Q50. [new]

Raw comma-separated lines arrive as strings. Parse each line into a dictionary (converting `gpa` from string to `float`), build a Pandas DataFrame, print the column data types, the mean GPA per major, and the top three students by GPA.

```
lines = ["Ali,DATA,3.4", "Sara,ICS,3.8", "Omar,DATA,2.9",
         "Huda,MATH,3.6", "Fahad,ICS,3.1"]
```

Answer:

```
import pandas as pd
records = []
for line in lines:
    name, major, gpa = line.split(",")
    records.append({"name": name, "major": major, "gpa": float(gpa)})
df = pd.DataFrame(records)
print(df.dtypes)                                  # gpa is float64
print(df.groupby("major")["gpa"].mean())          # DATA 3.15, ICS 3.45, MATH 3.60
print(df.sort_values("gpa", ascending=False).head(3))   # Sara, Huda, Ali
```

My answer